Decentralized Science on Zoo Network: Open Research Coordination with Verifiable Provenance

Infrastructure for Reproducible, Incentivized, and Community-Governed Science

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Abstract

Science is broken in ways that scientists have documented for decades: irreproducible results (>70% failure to replicate across fields), funding concentrated in a handful of institutions, data locked behind paywalls and proprietary silos, and credit attribution that rewards journal prestige over actual contribution. We present Zoo Network as infrastructure for **Decentralized Science (DeSci)**—a platform where research provenance is on-chain and immutable, computation is verifiable, data is open by default, funding flows through community governance, and mathematical claims carry machine-checked proofs. The system introduces **Research Capsules**: encrypted lab notebooks with threshold-gated data sharing and reproducible compute environments, anchored on I-Chain for tamper-evident provenance. **ZIPs (Zoo Improvement Proposals)** serve as the governance mechanism for research funding, replacing opaque grant committees with transparent community peer review and milestone-based disbursement from the DAO treasury via G-Chain. A **Data Commons** built on I-Chain DID-authenticated contributions tracks usage receipts and citation graphs, ensuring contributors receive credit and compensation. A-Chain provides AI-assisted research capabilities: automated literature review, hypothesis generation, and statistical analysis. For mathematical results, we integrate Lean 4 proof verification, creating a chain of formally verified claims that extends from theorem statement to on-chain publication. We demonstrate the platform through conservation science applications—satellite ecology, species classification, and habitat modeling—where Zoo's existing research infrastructure provides immediate, concrete value.

Keywords: decentralized science, DeSci, research provenance, reproducibility, verifiable computation, data commons, formal verification

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1 Introduction

1.1 The Reproducibility Crisis

In 2016, a Nature survey of 1,576 researchers found that more than 70% had failed to reproduce another scientist's experiments, and more than half had failed to reproduce their own [1]. This is not a fringe concern—it is a structural failure of the scientific enterprise.

The causes are well-understood:

1. **Publication Bias.** Journals publish positive results; negative results and replications languish unpublished. The literature is a biased sample of all experiments conducted.
2. **Methodological Opacity.** Papers describe methods in prose; the actual code, data, and parameters are often unavailable.
3. **Perverse Incentives.** Tenure, grants, and prestige are awarded for novel findings, not for careful replication.
4. **Data Unavailability.** Datasets are proprietary, deleted after publication, or stored in formats that cannot be re-processed.

1.2 The Funding Bottleneck

Scientific funding is concentrated, slow, and conservative:

- The NIH funds ∼20% of grant applications. The median PI age at first R01 grant is 43.
- Funding cycles are 12–18 months. By the time a grant is awarded, the research landscape has shifted.
- Geographic concentration: 80% of global R&D spending occurs in 10 countries.
- Interdisciplinary and high-risk research is systematically underfunded because review panels favor incremental advances in established fields.

1.3 Zoo as Research Infrastructure

Zoo Network addresses these failures not through exhortation but through infrastructure. The thesis is simple: **if the tools enforce reproducibility, transparency, and fair attribution, scientists will use them because they make science easier, not harder.**

Zoo provides:

1. **On-Chain Provenance.** Every dataset, experiment, and publication has an immutable timestamp and authorship record on I-Chain.
2. **Verifiable Computation.** Experiments run in containerized environments; execution traces are hashed and anchored on-chain.
3. **Open Data Commons.** DID-authenticated data contributions with usage receipts and citation tracking.
4. **Community Funding.** ZIPs and DAO governance replace opaque grant committees with transparent, milestone-based funding.

5. **AI-Assisted Research.** A-Chain inference for literature review, hypothesis generation, and automated analysis.

6. **Formal Verification.** Lean 4 proof chains for mathematical results.

1.4 Paper Organization

Section 2 introduces Research Capsules. Section 3 describes ZIPs as research proposals. Section 4 presents the Data Commons. Section 5 covers AI-assisted research. Section 6 demonstrates conservation science applications. Section 7 details formal verification. Section 8 analyzes platform economics. Section 9 surveys related work. Section 10 concludes.

2 Research Capsules

2.1 Encrypted Lab Notebooks

A Research Capsule is an encrypted, versioned, on-chain-anchored container for a complete research project. It extends the Memory Capsule architecture [4] with research-specific semantics:

Definition 1 (Research Capsule)**.** A Research Capsule $\mathcal{R} = (did, notebook, data, code, env, provenance)$ where:

- *did*: researcher's decentralized identifier on I-Chain.
- *notebook*: encrypted append-only log of observations, hypotheses, and analysis notes.
- *data*: encrypted datasets with content-addressed hashes.
- *code*: versioned source code (git-compatible).
- *env*: reproducible compute environment specification (container image hash).
- *provenance*: on-chain record of creation time, modifications, and access history.

2.2 Provenance Anchoring

Every mutation to a Research Capsule produces an on-chain anchor on I-Chain:

```
struct ProvenanceRecord {
    capsule_id:  bytes32,   // BLAKE3(capsule)
    author_did:  bytes32,   // researcher DID
    action:      uint8,     // 0=create, 1=update_data,
                            // 2=update_code, 3=run_experiment,
                            // 4=publish_result
    content_hash: bytes32,  // BLAKE3(changed_content)
    prev_hash:   bytes32,   // previous provenance record
    timestamp:   uint64     // block timestamp
}
```

This creates an immutable, publicly auditable timeline of the research process. Pre-registration of hypotheses (action 0 followed by action 3) is verifiable: the hypothesis was recorded before the experiment was run.

2.3 Threshold-Gated Data Sharing

Research data often has sensitivity constraints: patient data, endangered species locations, proprietary measurements. Research Capsules support threshold-gated sharing via T-Chain:

1. The researcher encrypts sensitive data under a threshold scheme (t-of-n via T-Chain).
2. Access requests require: (a) a valid DID with researcher credentials, (b) IRB approval credential (for human subjects data), and (c) t validator approvals.
3. Approved access produces a time-limited, scope-limited decryption key.
4. All access is logged on I-Chain for audit.

2.4 Reproducible Compute Environments

Each Research Capsule specifies its compute environment as an OCI container image:

- The image hash is recorded in the capsule's *env* field.
- Experiments are run inside the container, ensuring identical environments across reproductions.
- The execution trace (stdin/stdout/stderr + exit code + resource usage) is hashed and anchored on I-Chain.
- A reproduction attempt is valid if and only if: same container image, same input data hash, and matching output hash.

Definition 2 (Reproducible Experiment)**.** An experiment $E = (env, data, code, params)$ is *reproduced* by execution E' if and only if:

$$\text{BLAKE3}(output(E')) = \text{BLAKE3}(output(E)) \tag{1}$$

given $env(E') = env(E)$, $data(E') = data(E)$, $code(E') = code(E)$, and $params(E') = params(E)$.

For stochastic experiments, the definition relaxes to statistical equivalence: output distributions match within a specified tolerance.

3 ZIPs as Research Proposals

3.1 From Improvement Proposals to Research Proposals

Zoo Improvement Proposals (ZIPs) [5] are the governance mechanism for the Zoo Network. We extend ZIPs to serve as research funding proposals, replacing traditional grant applications:

Definition 3 (Research ZIP)**.** A Research ZIP $Z = (title, abstract, team, budget, milestones, timeline, capsule\_id)$ is a proposal for community-funded research, submitted to G-Chain for DAO vote.

3.2 Proposal Lifecycle

1. **Submission.** Researcher submits a Research ZIP to G-Chain with a deposit of 10M Keeper tokens (refundable upon completion).
2. **Community Review.** A 14-day review period where any Keeper holder can comment, ask questions, and signal support or opposition.
3. **Peer Review.** Three reviewers are selected from the reviewer pool (staked researchers with verified credentials on I-Chain). Reviewers stake 1M Keeper on their review quality.
4. **Vote.** A 7-day voting period on G-Chain. Standard proposals require 51% majority with 5% quorum; large grants (>100M Keeper) require 66% supermajority with 10% quorum.
5. **Funding.** Approved proposals receive milestone-based funding from the DAO treasury.
6. **Milestones.** Each milestone requires a deliverable (data, code, paper) committed to the Research Capsule. Milestone completion is verified by the original reviewers.
7. **Completion.** Final milestone triggers deposit refund and completion badge on the researcher's I-Chain DID.

3.3 Incentivized Peer Review

Peer review on Zoo is not volunteer labor—it is staked, compensated, and reputation-building:

- **Reviewer Stake.** Reviewers stake 1M Keeper when accepting a review assignment. The stake is slashed if the review is flagged as low-quality by the community (>10% of voters object).
- **Compensation.** Reviewers earn 500K Keeper per completed review, paid from the proposal's budget.
- **Reputation.** Completed reviews accrue reputation tokens on the reviewer's DID. High-reputation reviewers are prioritized for future assignments.
- **Conflict Checks.** I-Chain DID credentials are used to verify that reviewers have no institutional or co-authorship conflicts with the proposer.

3.4 Advantages Over Traditional Grants

4 Data Commons

4.1 DID-Authenticated Data Contributions

The Zoo Data Commons [6] is a decentralized repository where researchers contribute datasets with cryptographic provenance:

1. The contributor authenticates via their I-Chain DID.
2. The dataset is hashed (BLAKE3) and the hash is anchored on I-Chain with the contributor's DID as author.

Property	Traditional Grants	Research ZIPs
Decision timeline	12–18 months	21 days
Reviewer compensation	$0	500K Keeper
Review transparency	Opaque	Public
Funding source	Government/foundation	Community DAO
Geographic restriction	Yes (citizenship, residency)	No
Progress tracking	Annual reports	On-chain milestones
Reproducibility requirement	Rare	Mandatory

Table 1: Research ZIPs vs. traditional grant funding

3. The dataset is stored on decentralized storage (IPFS/Filecoin) with the on-chain hash as the content address.
4. Metadata (schema, license, domain, quality metrics) is stored on-chain.

4.2 Usage Receipts

Every use of a dataset in a Research Capsule generates a *usage receipt*—an on-chain record linking the consuming capsule to the contributing dataset:

```
struct UsageReceipt {
    dataset_hash:  bytes32,
    consumer_did:  bytes32,
    capsule_id:    bytes32,
    usage_type:    uint8,    // 0=training, 1=validation,
                             // 2=analysis, 3=reference
    timestamp:     uint64
}
```

Usage receipts serve two purposes:

1. **Attribution.** Data contributors receive verifiable credit for each use of their data.
2. **Compensation.** Contributors earn Keeper tokens proportional to usage. The default rate is 100 Keeper per receipt, paid from the consuming capsule's budget.

4.3 Citation Graphs on G-Chain

Research Capsules cite each other, creating a directed acyclic graph of scientific knowledge. This citation graph is stored on G-Chain and provides:

- **Impact Metrics.** Citation count, h-index, and PageRank-style importance scores computed on-chain.
- **Retroactive Funding.** High-impact work (measured by citations) can receive retroactive DAO funding, rewarding quality after the fact rather than guessing impact before.
- **Dependency Tracking.** If a foundational dataset or method is retracted, all dependent capsules are flagged automatically.

4.4 FAIR Compliance

The Data Commons natively satisfies the FAIR principles [2]:

- **Findable.** Content-addressed with persistent identifiers on I-Chain.
- **Accessible.** Open by default. Sensitive data is threshold-gated but discoverable (metadata is public).
- **Interoperable.** Schema metadata enforces standard formats. Cross-references use DIDs.
- **Reusable.** License and provenance are on-chain. Usage terms are machine-readable.

5 AI-Assisted Research

5.1 A-Chain Inference for Research

Zoo A-Chain provides AI inference as a network service. For DeSci, this enables:

5.1.1 Automated Literature Review

A researcher submits a topic query to A-Chain. The inference engine:

1. Searches the Data Commons for relevant capsules (using embeddings stored on-chain).
2. Retrieves and summarizes relevant findings from public Research Capsules.
3. Generates a structured literature review with proper citations (linked to on-chain capsule IDs).
4. Identifies gaps: topics with high query volume but few capsules.

Cost: 500 Keeper per literature review (inference credits).

5.1.2 Hypothesis Generation

Given a dataset and a research question, A-Chain inference can:

1. Analyze the dataset's statistical properties.
2. Cross-reference with existing findings in the Data Commons.
3. Propose testable hypotheses ranked by novelty and feasibility.
4. Pre-register proposed hypotheses on I-Chain (timestamped before experimentation).

5.1.3 Automated Statistical Analysis

A-Chain provides standardized statistical analysis pipelines:

- Power analysis for sample size determination.
- Appropriate test selection based on data distribution and research design.
- Multiple comparison correction (Bonferroni, FDR).
- Effect size estimation with confidence intervals.
- Automated reporting in standardized format.

5.2 Privacy-Preserving AI Analysis

For sensitive data (medical records, endangered species locations), AI analysis runs on encrypted data via CKKS on A-Chain [12]:

1. The researcher encrypts the dataset under the collective FHE key.
2. A-Chain runs the analysis model on encrypted inputs.
3. Results are returned encrypted; only the researcher can decrypt.
4. The analysis trace (encrypted) is anchored on I-Chain for reproducibility.

This enables AI-assisted analysis of sensitive data without the AI model (or its operators) ever observing the plaintext.

6 Conservation Science Applications

Zoo Labs Foundation's mission centers on conservation. The DeSci platform provides immediate value through three existing research programs.

6.1 Satellite Ecology

The Zoo satellite ecology program [8] uses remote sensing data for habitat monitoring:

- **Data.** Sentinel-2 and Landsat imagery contributed to the Data Commons by governmental and institutional partners.
- **Analysis.** A-Chain runs land-use classification models on satellite imagery, detecting deforestation, habitat fragmentation, and urban encroachment.
- **Provenance.** Analysis results are anchored in Research Capsules with full reproducibility (container image + input data hash + model weights hash).
- **Funding.** Satellite ecology research is funded through Research ZIPs, with milestone deliverables including classified imagery and habitat change reports.

6.2 Species Classification

The species classification pipeline [9] uses AI for biodiversity assessment:

- **Data.** Camera trap images, acoustic recordings, and citizen-science observations contributed to the Data Commons.
- **Models.** Classification models trained on A-Chain with federated learning across institutions [10]. Each institution contributes training data without sharing raw images (privacy preserved via federated aggregation).
- **Validation.** Expert validation of model predictions is incentivized: domain experts earn Keeper for confirming or correcting classifications.
- **Output.** Species distribution maps, population estimates, and trend analyses published as Research Capsules.

6.3 Habitat Modeling

Predictive habitat modeling integrates satellite, species, and climate data:

- **Multi-Source Fusion.** A-Chain inference combines satellite-derived land cover, species occurrence records, and climate projections.
- **Scenario Analysis.** Models evaluate habitat viability under different climate scenarios (RCP 2.6, 4.5, 8.5).
- **Conservation Planning.** Output feeds directly into conservation action plans, identifying priority areas for protection.
- **Impact Bonds.** Conservation outcomes (measured via satellite monitoring) can trigger Zoo impact bond payouts [11], creating a financial feedback loop between science and action.

7 Formal Verification of Research Claims

7.1 The Trust Problem in Mathematical Results

Published mathematical results are trusted because peer reviewers checked them. But peer review is fallible: errors in published proofs are discovered regularly, sometimes decades later. For results that underpin critical infrastructure (cryptographic protocols, safety-critical systems), informal trust is insufficient.

7.2 Lean 4 Proof Chain

Zoo integrates Lean 4 [3] formal verification into the research publication pipeline:

1. The researcher states a theorem in Lean 4.
2. The proof is developed interactively, with each step machine-checked.
3. The completed proof is compiled to a proof certificate.
4. The certificate hash is anchored on I-Chain alongside the Research Capsule.
5. Any verifier can download the proof and re-check it independently.

7.3 Proof-Linked Publications

A Research Capsule containing a formal proof links the following chain:

$$\textit{theorem} \xrightarrow{\text{Lean 4}} \textit{proof} \xrightarrow{\text{BLAKE3}} \textit{hash} \xrightarrow{\text{IChain}} \textit{anchor} \tag{2}$$

This creates an immutable, machine-verified record that the theorem was proved (not merely claimed) at the time of publication.

7.4 Verified Cryptographic Claims

For Zoo's own cryptographic infrastructure, formal verification provides a trust foundation:

- CKKS rescaling correctness (`Crypto/FHE/CKKS.lean`) [12]
- TFHE noise budget tracking (`Crypto/FHE/TFHE.lean`)
- Threshold signature security (`Crypto/Threshold/FROST.lean`)
- GPU scaling bounds (`GPU/FHEScaling.lean`)

These proofs are anchored on I-Chain and linked from the corresponding technical papers, allowing anyone to verify that the security claims are not merely argued but proved.

7.5 Incentivizing Formalization

Formalizing existing results is valuable but tedious. Zoo incentivizes it:

- **Formalization Bounties.** DAO treasury funds bounties for formalizing high-impact results.
- **Reputation Rewards.** Formalization efforts accrue reputation on the researcher's DID.
- **Citation Credit.** A formalized proof cites the original result, generating a usage receipt and compensation for the original authors.

8 Platform Economics

8.1 Revenue Flows

Activity	Revenue	Recipient
Data contribution (per usage receipt)	100 Keeper	Data contributor
Peer review (per review)	500K Keeper	Reviewer
AI inference (per query)	500 Keeper	Validator network
Capsule storage (per GB/month)	10K Keeper	Validator network
Formal verification bounty	Variable	Formalizer

Table 2: DeSci platform revenue flows

8.2 Sustainability Model

The DeSci platform is funded by the DAO treasury (1T Keeper allocated at genesis [7]). Sustainability depends on:

1. **Network Effects.** More data in the Commons attracts more researchers, who contribute more data.
2. **Usage Fees.** AI inference and storage fees generate ongoing revenue for validators.
3. **External Funding.** Institutional partners (universities, governments, NGOs) contribute Keeper to fund research in their domains.

4. **Impact Bonds.** Conservation outcomes trigger bond payouts, recycling capital back into the research pipeline.

8.3 Cost Comparison

Activity	Traditional Cost	Zoo Cost
Grant application	200+ person-hours	10 person-hours
Peer review cycle	6–12 months	21 days
Data hosting (per TB/year)	\$500–2,000	120M Keeper (∼\$240)
Reproducibility verification	Weeks (manual)	Minutes (automated)
Cross-institution collaboration	Legal agreements	DID capability tokens

Table 3: Cost comparison: traditional vs. Zoo DeSci infrastructure

9 Related Work

9.1 DeSci Projects

VitaDAO funds longevity research through token-governed grants. Unlike Zoo, VitaDAO is domain-specific (longevity) and does not provide research infrastructure (computation, data hosting, reproducibility). **ResearchHub** incentivizes open-access publishing with a token reward for peer review. It focuses on the publication layer, not the research process. **DeSci Labs** builds tools for research data management but without on-chain provenance or verifiable computation.

Zoo differs from all existing DeSci projects in providing a *complete* research infrastructure stack: from data contribution through computation, review, funding, publication, and formal verification.

9.2 Reproducibility Platforms

Code Ocean provides containerized compute capsules for reproducible research. **Binder** offers free ephemeral environments for Jupyter notebooks. **Whole Tale** integrates data, code, and compute for reproducibility.

These platforms provide reproducibility but lack: on-chain provenance, decentralized governance, economic incentives for contribution, and formal verification integration.

9.3 Open Data Initiatives

Zenodo (CERN) provides open-access data hosting with DOIs. **Dataverse** offers institutional data repositories. **GBIF** aggregates biodiversity occurrence records.

These initiatives provide data hosting but not: usage-based compensation, decentralized governance, or AI-assisted analysis.

10 Conclusion

The problems of science—irreproducibility, funding concentration, data silos, attribution failures—are infrastructure problems. They persist not because scientists are negligent but because the infrastructure makes negligence the path of least resistance.

Zoo Network does not appeal to better behavior. It changes the infrastructure. Research Capsules make reproducibility the default. ZIP-based funding eliminates geographic and institutional barriers. Usage receipts ensure contributors are compensated. Formal verification makes trust in mathematical claims unnecessary—the machine checks.

The conservation science applications demonstrate that this is not speculative. Satellite ecology, species classification, and habitat modeling are active research programs on Zoo Network, producing data and results today. The DeSci platform extends this foundation to all fields of science.

Two open questions remain:

1. **Adoption Incentives.** How to bootstrap the Data Commons past the critical mass where network effects take hold. Initial conservation datasets provide a nucleus; scaling to other domains requires domain-specific outreach.
2. **Regulatory Integration.** How on-chain research provenance integrates with existing regulatory frameworks (FDA, EMA) for research that informs policy or clinical practice.

Science deserves infrastructure as rigorous as the claims it produces. Zoo provides it.

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